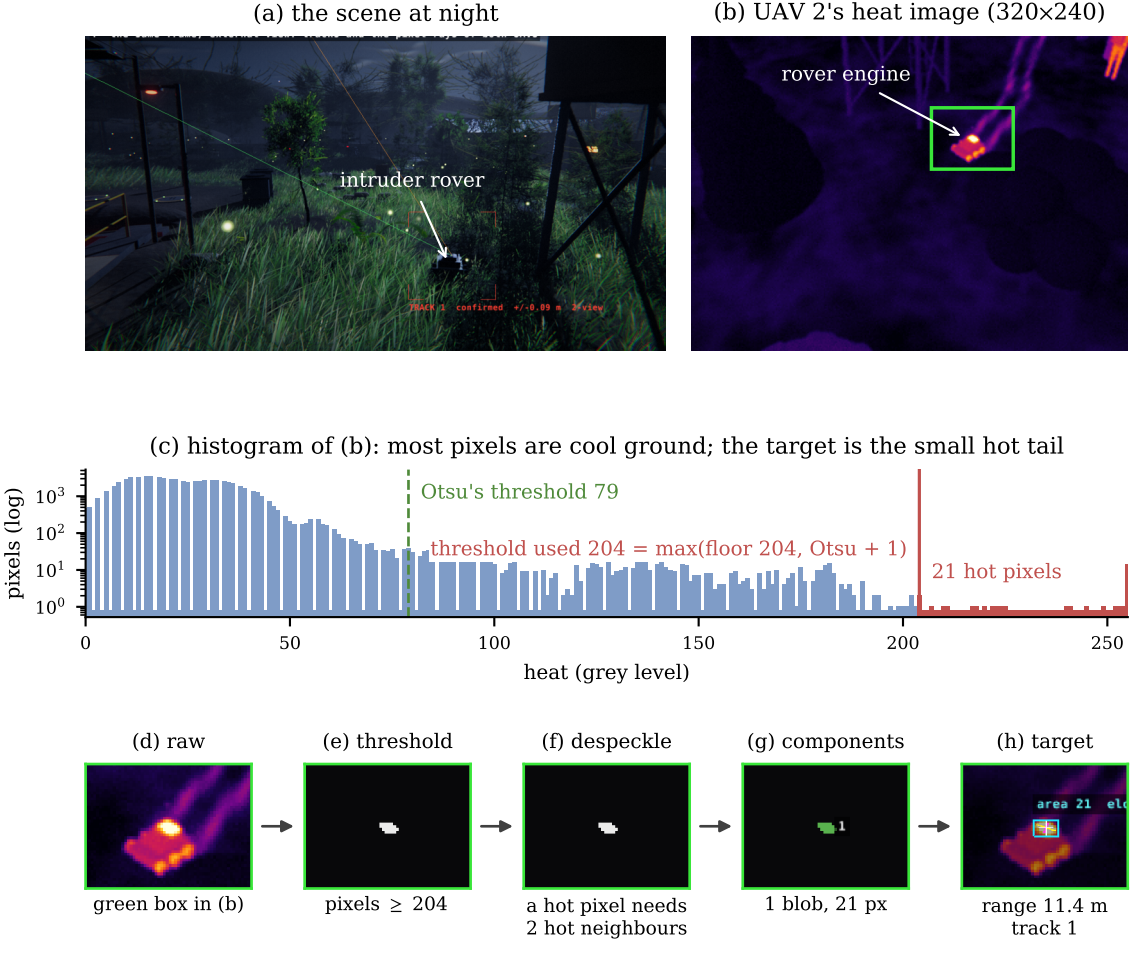

(a) the scene at night
intruder rover
(b) UAV 2's heat image (320×240)
rover engine
(c) histogram of (b): most pixels are cool ground; the target is the small hot tail
pixels (log)
10^3
10^2
10^1
10^0
Otsu's threshold 79
threshold used 204 = max(floor 204, Otsu + 1)
21 hot pixels
0
50
100
150
200
250
heat (grey level)
(d) raw
(e) threshold
(f) despeckle
(g) components
(h) target
area 21
1
green box in (b)
pixels ≥ 204
a hot pixel needs
2 hot neighbours
1 blob, 21 px
range 11.4 m
track 1